

DATE: EXPT NO:08	MINI PROJECT Social Media Analytics
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AIM:

To analyze airline passenger satisfaction data using Python, SQL Server, and Tableau by cleaning and managing passenger records, performing data analysis, and creating interactive visualizations to identify factors affecting customer satisfaction such as travel class, service quality, flight distance, delays, and onboard facilities.

Software and Tools Required

- Python
- VS Code
- Pandas
- SQL Server
- SQL Server Management Studio (SSMS)
- Tableau Desktop

3. Process / Procedure

Step 1 — Project Setup

Create a project folder:

AirlinePassengerAnalysis

|

|— .venv

|— analysis.py

|— insert_data.py

|— airline_passenger_satisfaction.csv

Create and activate the Python virtual environment.

Step 2 — Create Database

Open SQL Server Management Studio (SSMS) and connect using:

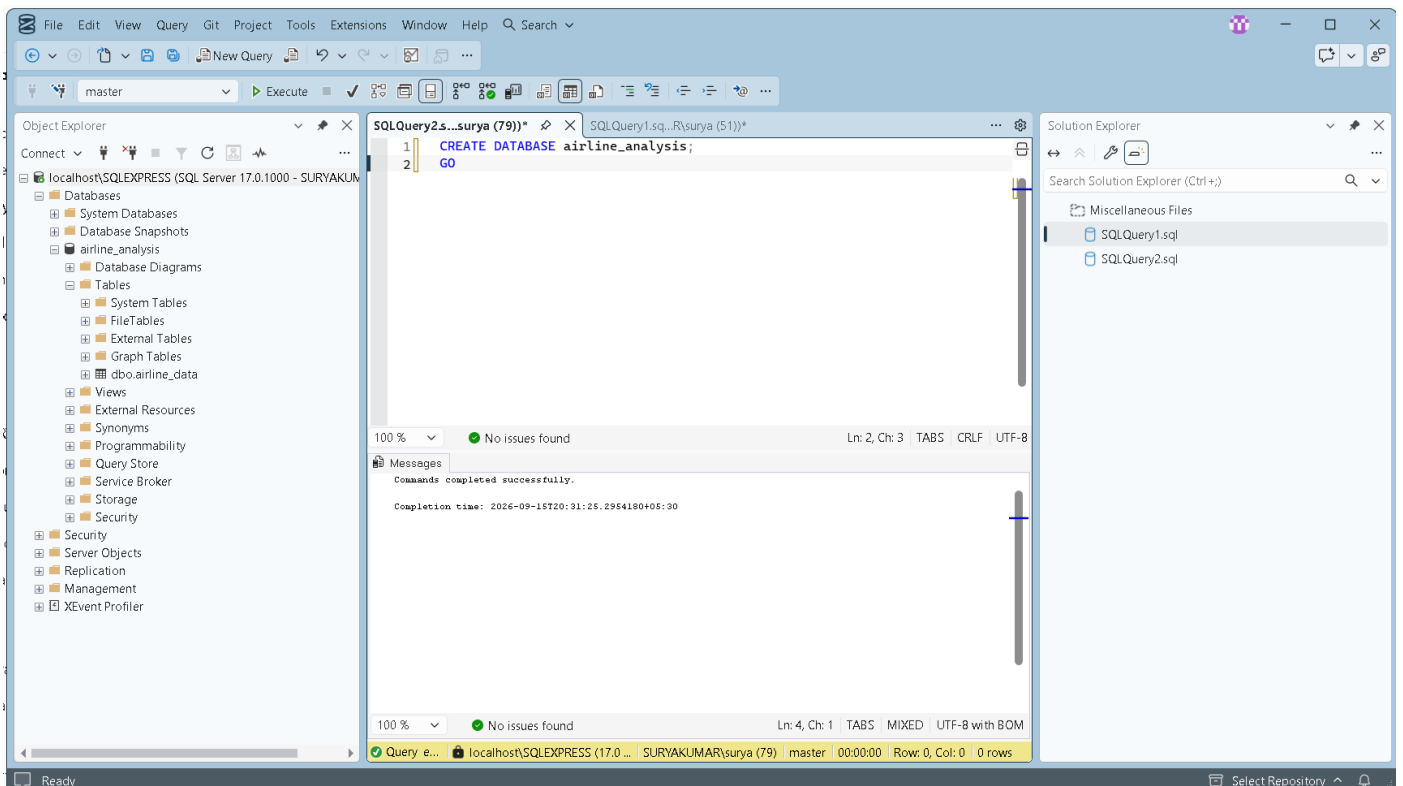
Server: localhost

Authentication: Windows Authentication

Create the database:

```
CREATE DATABASE airline_analysis;  
GO
```

OUTPUT:



Step 3 — Create Table

Create the **airline_data** table:

```
USE airline_analysis;  
GO
```

```
CREATE TABLE airline_data (  
    Passenger_ID INT,  
    Gender VARCHAR(20),  
    Customer_Type VARCHAR(50),  
    Age INT,
```

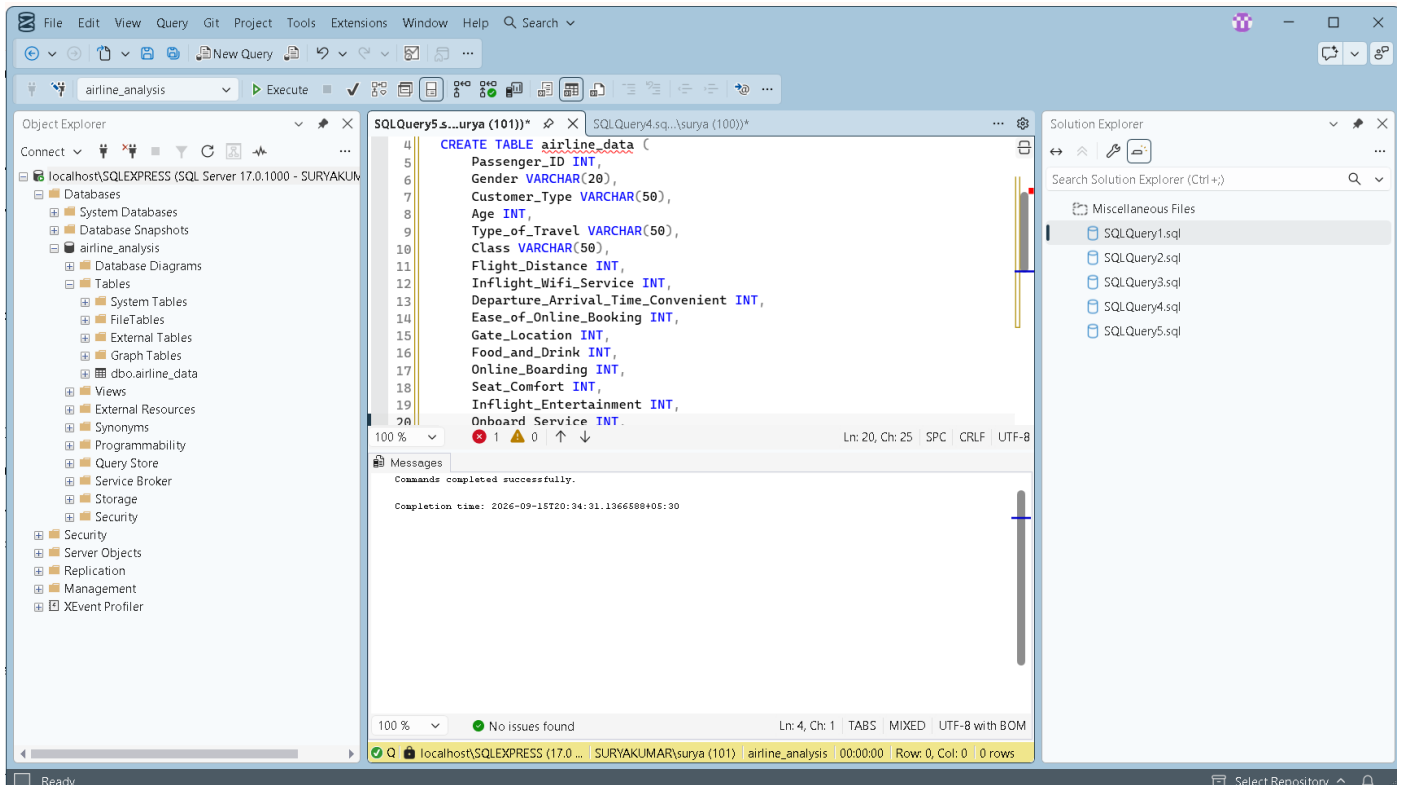
**Type_of_Travel VARCHAR(50),
Class VARCHAR(50),
Flight_Distance INT,**

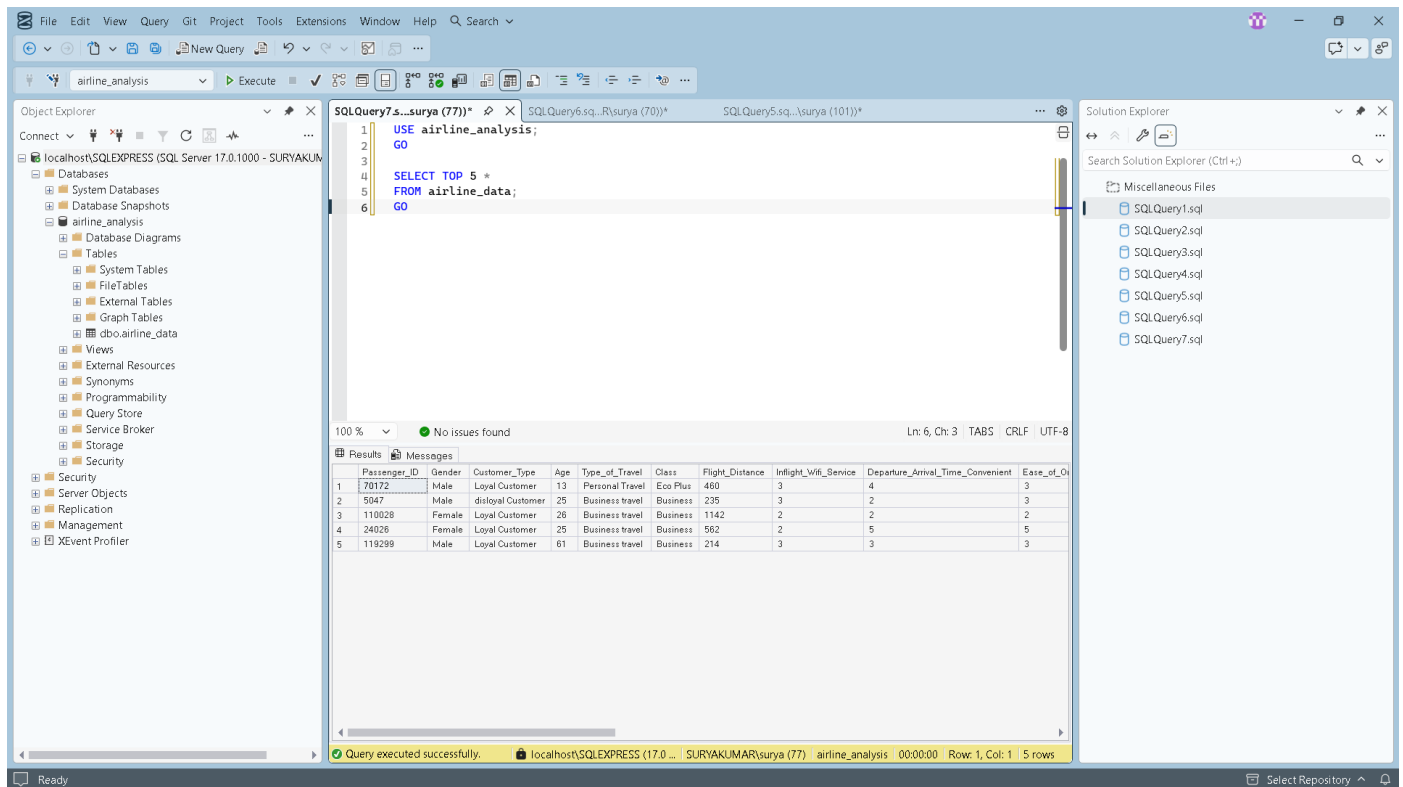
**Inflight_Wifi_Service INT,
Departure_Arrival_Time_Convenient INT,
Ease_of_Online_Booking INT,
Gate_Location INT,
Food_and_Drink INT,
Online_Boarding INT,
Seat_Comfort INT,
Inflight_Entertainment INT,
Onboard_Service INT,
Leg_Room_Service INT,
Baggage_Handling INT,
Checkin_Service INT,
Inflight_Service INT,
Cleanliness INT,
Departure_Delay INT,
Arrival_Delay FLOAT,
Satisfaction VARCHAR(50)**

);

GO

OUTPUT:





Step 4 — Load and Analyze Dataset Using Python

Program 1 — Load Dataset and View Information

Code

```
import pandas as pd

# Load dataset
df = pd.read_csv("airline_passenger_satisfaction.csv")

# Show first 5 rows
print("FIRST 5 ROWS:")
print(df.head())

print("\n")

# Dataset information
print("DATASET INFO:")
print(df.info())
```

Purpose

Load the dataset and display the first five records, column names, data types, and dataset information.

Output

```
PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> & C:\Python314\python.exe c:/Users/surya/OneDrive/Desktop/AirlinePassengerAnalysis/analysis.py
FIRST 5 ROWS:
   Unnamed: 0  ...      satisfaction
0           0  ...  neutral or dissatisfied
1           1  ...  neutral or dissatisfied
2           2  ...      satisfied
3           3  ...  neutral or dissatisfied
4           4  ...      satisfied

[5 rows x 25 columns]
```

Program 2 — Check Missing Values

Code

```
import pandas as pd

# Load dataset
df = pd.read_csv("airline_passenger_satisfaction.csv")

print("MISSING VALUES:")
print(df.isnull().sum())
```

Purpose

Check for missing values in each column of the dataset.

Output

```
PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> & C:\Python314\python.exe c:/Users/surya/OneDrive/Desktop/AirlinePassengerAnalysis/analysis.py
MISSING VALUES:
   Unnamed: 0      0
   id         0
   Gender      0
   Customer Type  0
   Age         0
   Type of Travel  0
   Class       0
   Flight Distance  0
   Inflight wifi service  0
   Departure/Arrival time convenient  0
   Ease of Online booking  0
   Gate location      0
   Food and drink     0
   Online boarding    0
   Seat comfort       0
   Inflight entertainment  0
   On-board service   0
   Leg room service   0
   Baggage handling   0
   Checkin service    0
   Inflight service   0
   Cleanliness        0
   Departure Delay in Minutes  0
   Arrival Delay in Minutes  310
   satisfaction       0
dtype: int64
```

Program 3 — Data Cleaning

Code

```
import pandas as pd

# Load dataset
df = pd.read_csv("airline_passenger_satisfaction.csv")

print("Before Cleaning:")
print(df["Arrival Delay in Minutes"].isnull().sum())

# Fill missing values with mean
df["Arrival Delay in Minutes"] = df["Arrival Delay in Minutes"].fillna(
    df["Arrival Delay in Minutes"].mean()
)

print("\nAfter Cleaning:")
print(df["Arrival Delay in Minutes"].isnull().sum())

# Save cleaned dataset
df.to_csv("cleaned_airline_data.csv", index=False)

print("\nCleaned dataset saved successfully!")
```

Purpose

Replace missing values in "Arrival Delay in Minutes" with the mean value and save the cleaned dataset.

Output

```
C:\Users\surya> cd C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis & python c:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis\analysis.py
PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> & C:\Python314\python.exe c:/Users/surya/OneDrive/Desktop/AirlinePassengerAnalysis/analysis.py
Before Cleaning:
310

After Cleaning:
0
```

Summary

Dataset Name: Airline Passenger Satisfaction Dataset

Total Records: 103,904

Total Columns: 25

Missing Values Found: 310

Missing Values After Cleaning: 0

Output File Created: cleaned_airline_data.csv

Step 5 — Insert Data into SQL Server

The Python `insert_data.py` program connects to SQL Server and inserts the cleaned dataset into the database.

The data flow is:

Python

↓

`cleaned_airline_data.csv`

↓

SQL Server

↓

`airline_analysis`

↓

`airline_data`

CODE:

```
import pandas as pd
import pyodbc

# Connect to SQL Server
conn = pyodbc.connect(
    r"DRIVER={SQL Server};"
    r"SERVER=localhost\SQLEXPRESS;"
    r"DATABASE=airline_analysis;"
    r"Trusted_Connection=yes;"
)

cursor = conn.cursor()

# Load cleaned CSV
df = pd.read_csv("cleaned_airline_data.csv")

# Insert rows
for index, row in df.iterrows():

    cursor.execute("""
INSERT INTO airline_data VALUES (
    ?,?,?,?,?,?,?,?,
    ?,?,?,?,?,?,?,?,
    ?,?,?,?
    )
    """)
```

```

"""
    int(row['id']),
    row['Gender'],
    row['Customer Type'],
    int(row['Age']),
    row['Type of Travel'],
    row['Class'],
    int(row['Flight Distance']),
    int(row['Inflight wifi service']),
    int(row['Departure/Arrival time convenient']),
    int(row['Ease of Online booking']),
    int(row['Gate location']),
    int(row['Food and drink']),
    int(row['Online boarding']),
    int(row['Seat comfort']),
    int(row['Inflight entertainment']),
    int(row['On-board service']),
    int(row['Leg room service']),
    int(row['Baggage handling']),
    int(row['Checkin service']),
    int(row['Inflight service']),
    int(row['Cleanliness']),
    int(row['Departure Delay in Minutes']),
    float(row['Arrival Delay in Minutes']),
    row['satisfaction']
)

conn.commit()

print("Data inserted successfully!")

cursor.close()
conn.close()

```

OUTPUT:

```

PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> pip install pyodbc
Defaulting to user installation because normal site-packages is not writeable
Collecting pyodbc
  Downloading pyodbc-5.3.0-cp314-cp314-win_amd64.whl.metadata (2.8 kB)
  Downloading pyodbc-5.3.0-cp314-cp314-win_amd64.whl (72 kB)
Installing collected packages: pyodbc
Successfully installed pyodbc-5.3.0
PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> python insert_data.py
Data inserted successfully!
PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> 

```

Step 6 — Verify the Dataset

Run:

**USE airline_analysis;
GO**

**SELECT COUNT(*) AS total_records
FROM airline_data;**

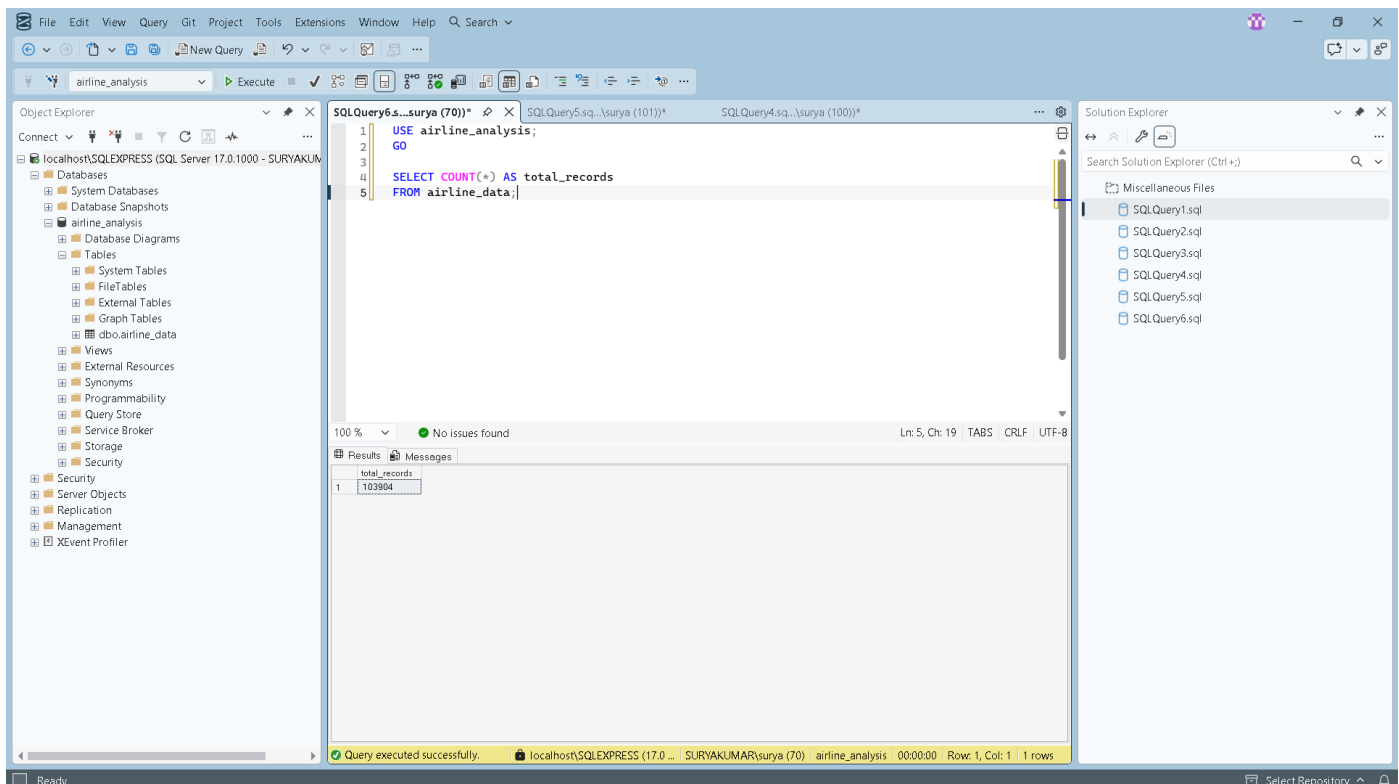
Result:

total_records

103904

This confirms that all passenger records have been successfully stored in SQL Server.

OUTPUT



3. Data Visualization Using Power BI

After storing the dataset in SQL Server, connect Power BI Desktop to:

SQL Server



airline_analysis



airline_data

Create different visualizations to analyze passenger satisfaction and airline service quality.

Visualization 1 — Satisfaction Distribution

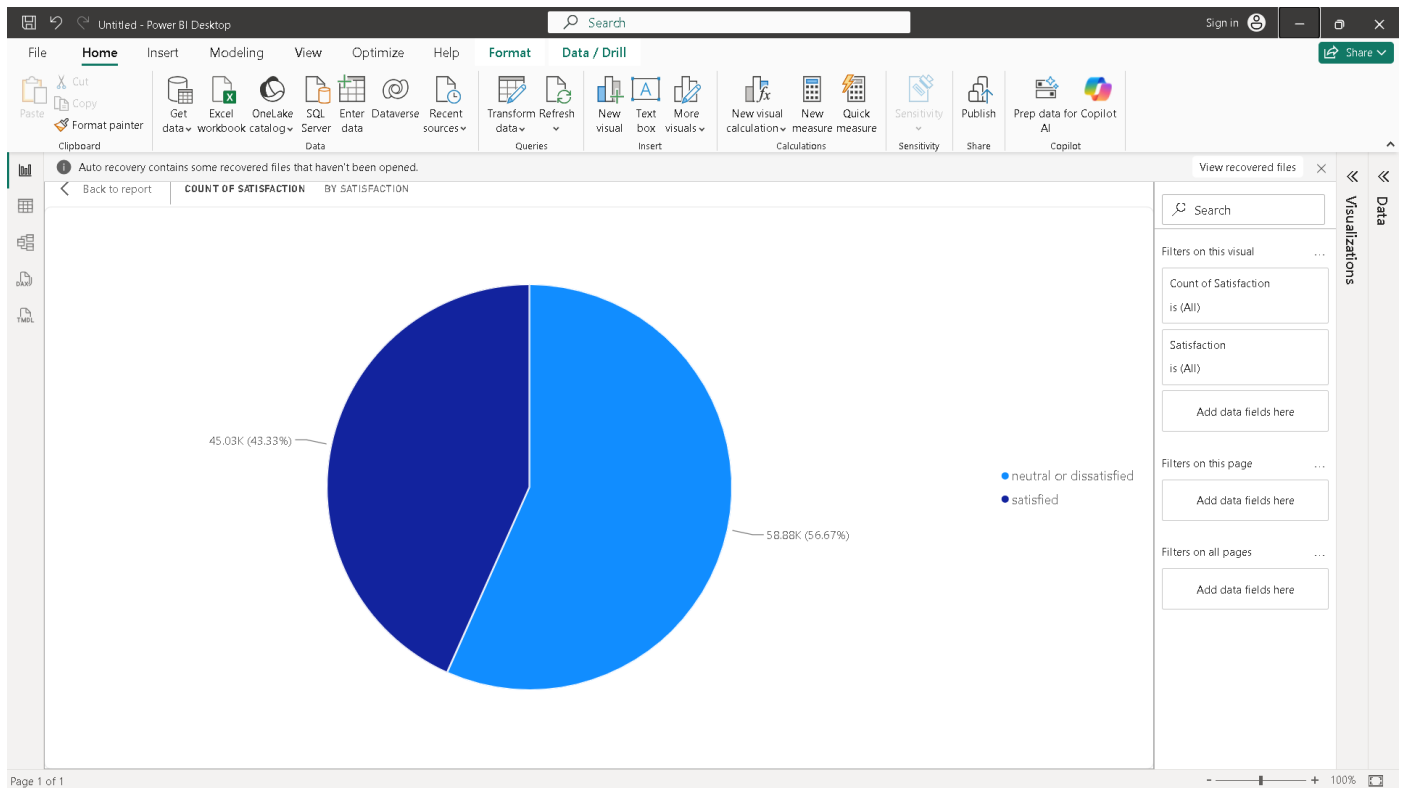
Chart: Pie Chart

Shows the distribution of passengers based on satisfaction level.

Satisfied

Neutral or Dissatisfied

Purpose: To identify the overall passenger satisfaction percentage.



Visualization 2 — Class-wise Satisfaction Analysis

Chart: Bar Chart

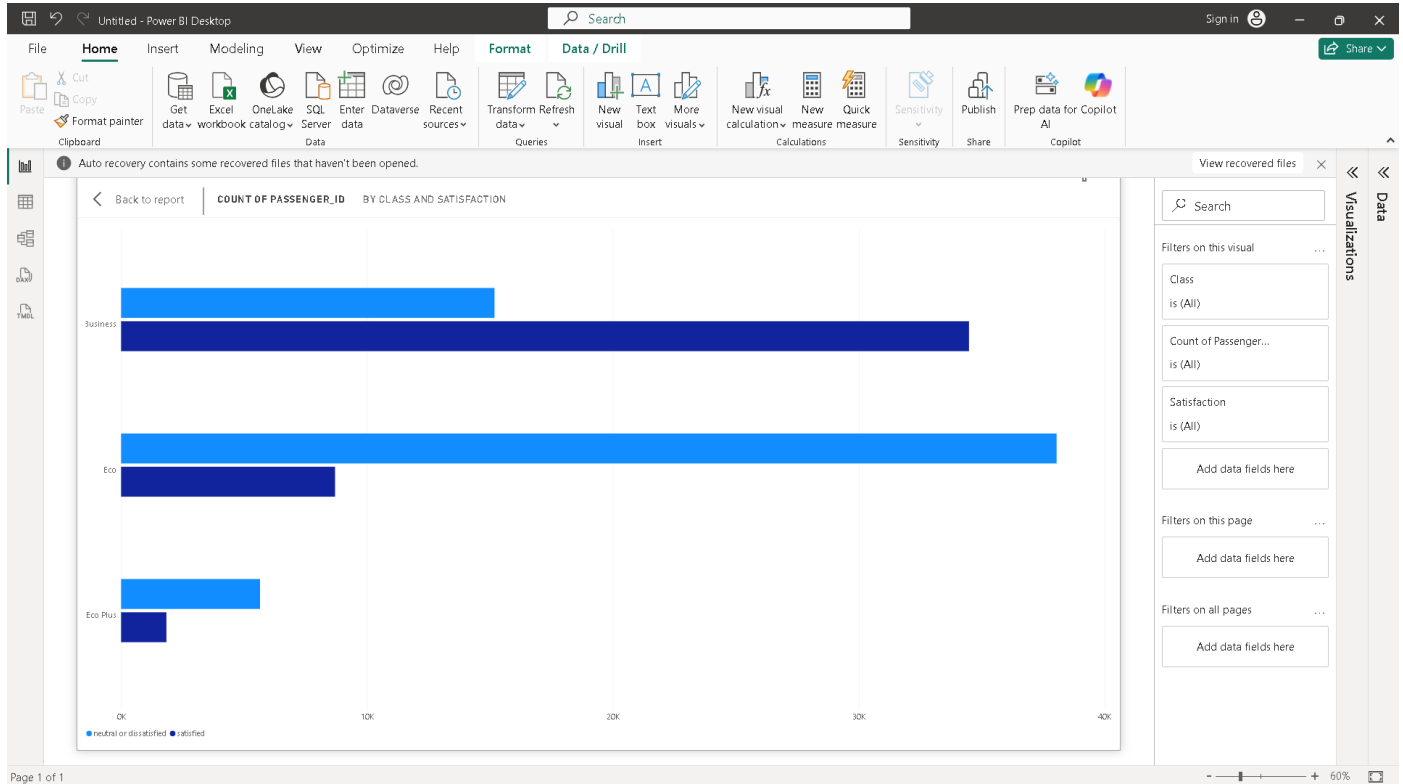
Shows passenger satisfaction across different travel classes.

Business

Eco

Eco Plus

Purpose: To analyze how travel class affects passenger satisfaction.



Visualization 3 — Gender Distribution

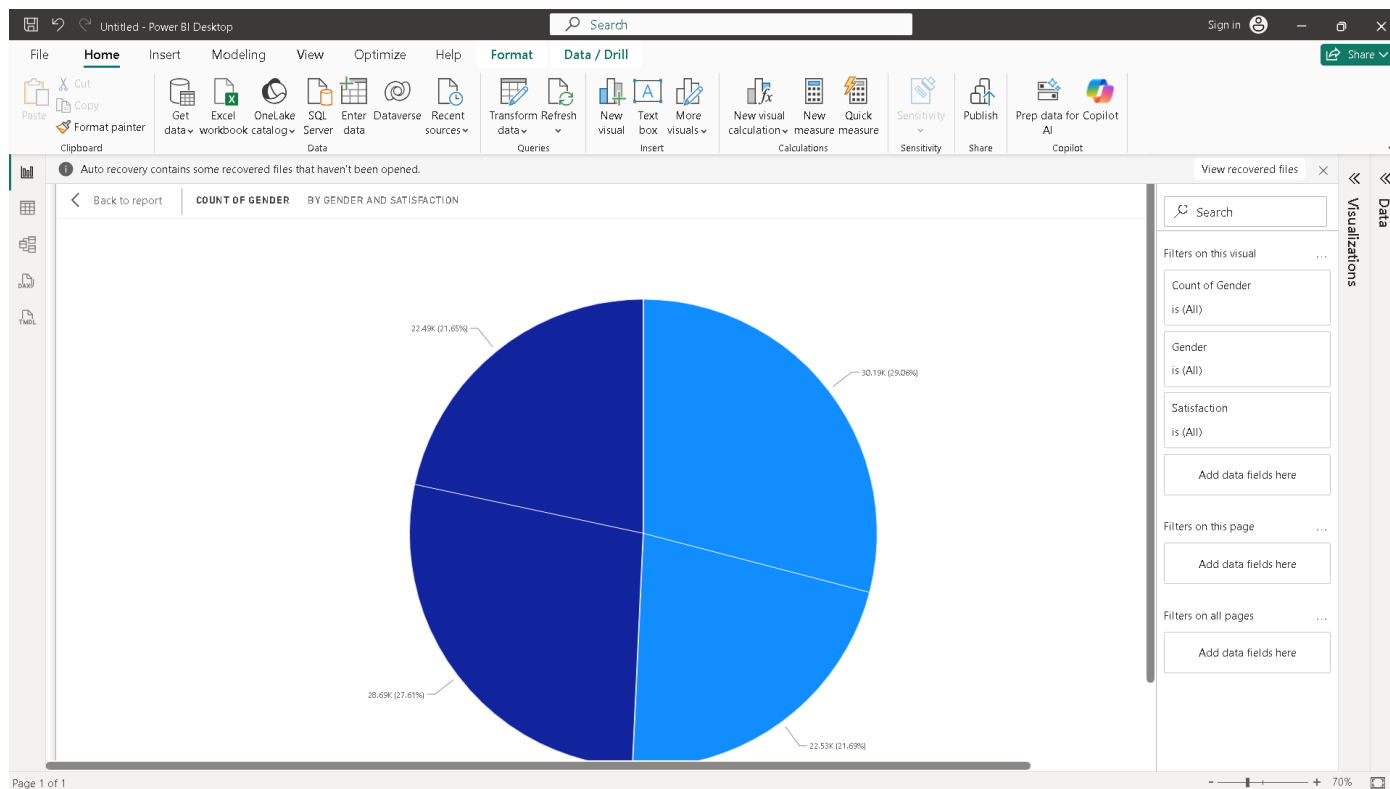
Chart: Pie Chart

Shows the distribution of passengers by gender.

Male

Female

Purpose: To understand the passenger demographic distribution.

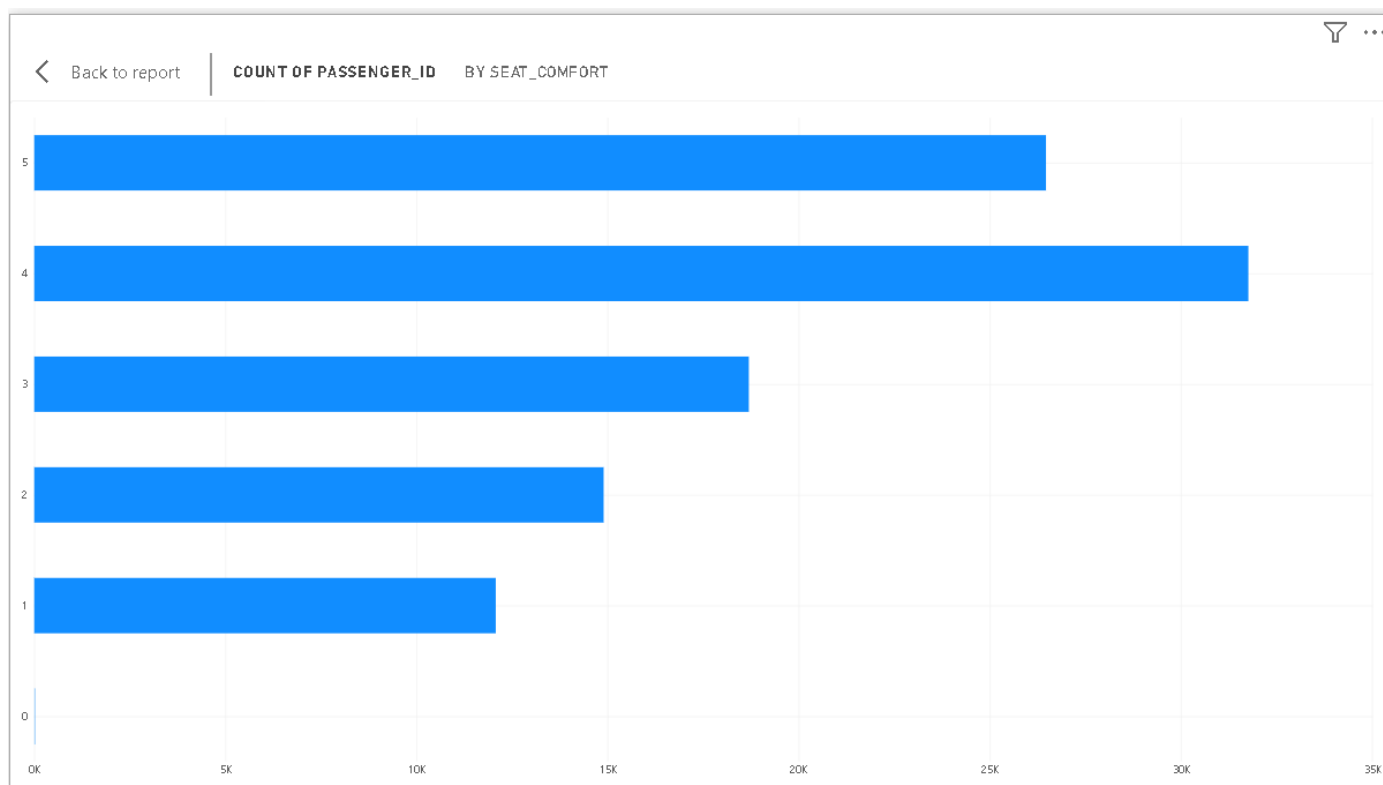


Visualization 5 — Seat Comfort vs Satisfaction

Chart: Bar Chart

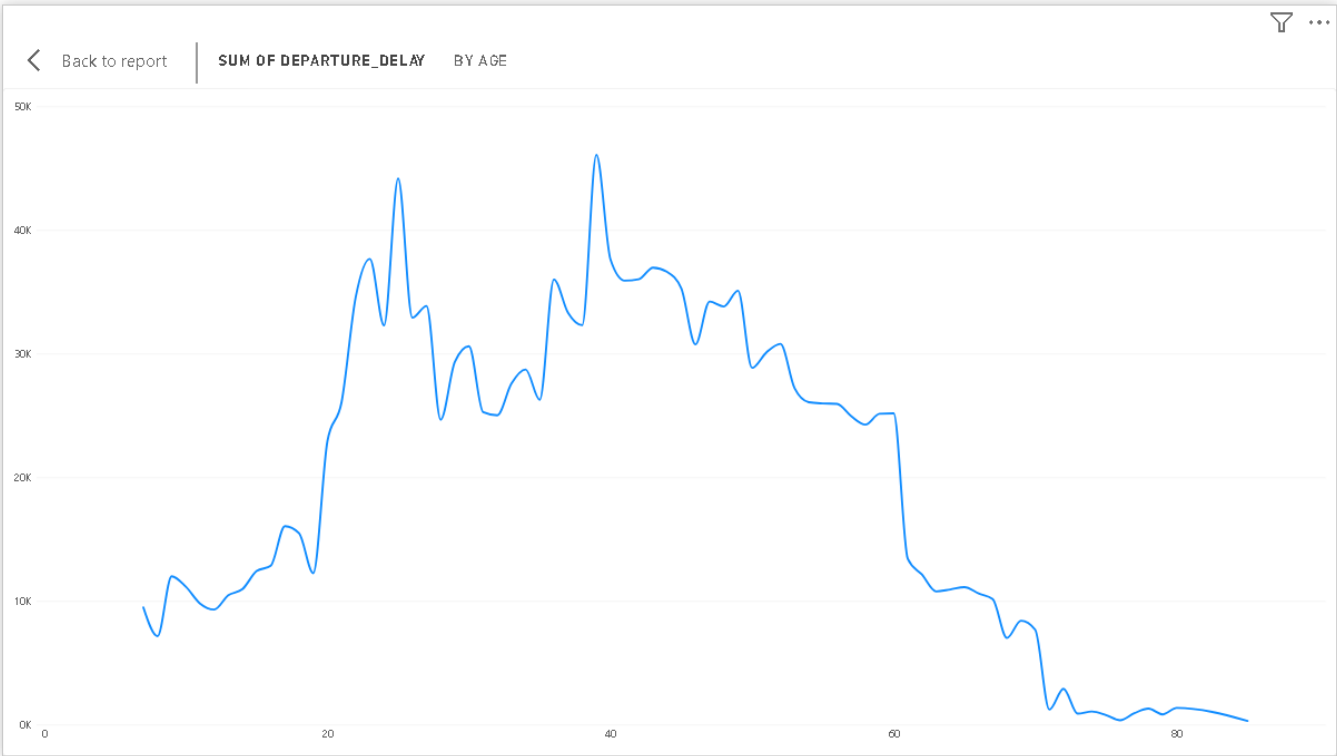
Compares average seat comfort ratings with passenger satisfaction levels.

Purpose: To determine the impact of seat comfort on customer satisfaction.



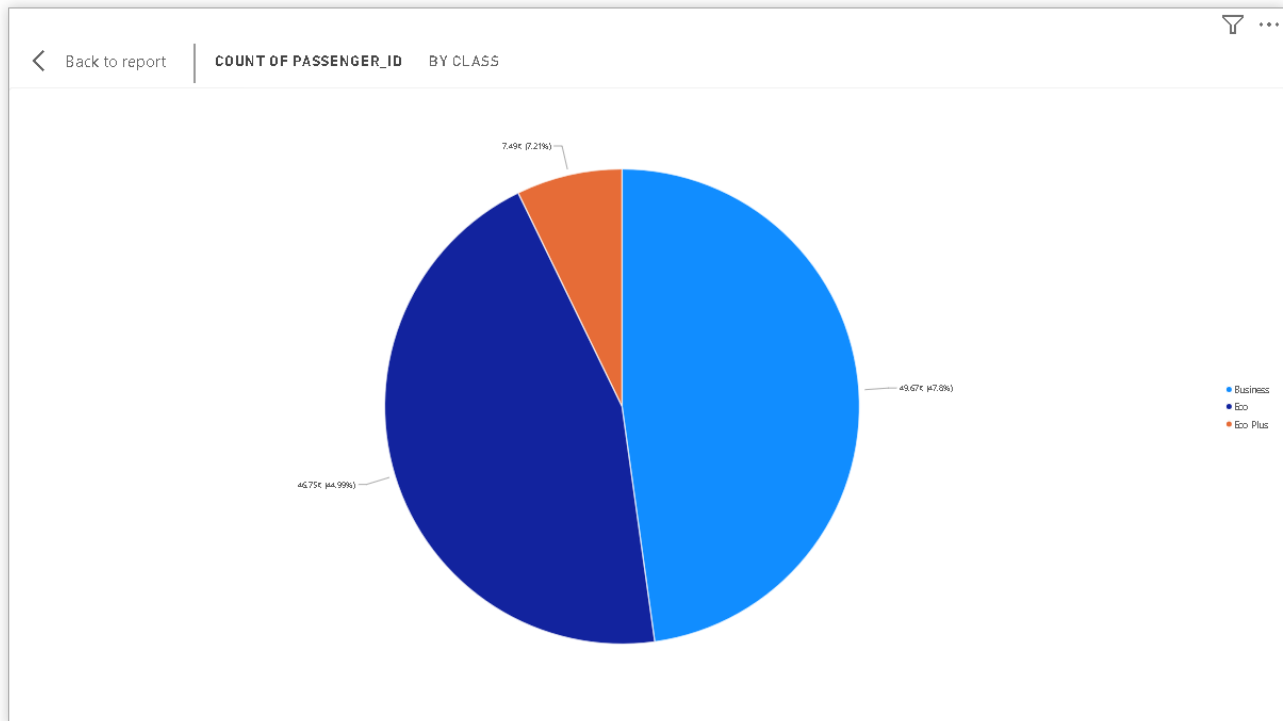
Visualization 7 — Delay Analysis

Chart: Line Chart
Shows departure and arrival delays.
Departure Delay
Arrival Delay
Purpose: To identify delay patterns and their impact on passenger experience.



Visualization 8 — Passenger Distribution by Travel Class

Chart: Pie Chart
This visualization shows the distribution of passengers across different travel classes such as Business, Eco, and Eco Plus. It helps analyze the proportion of passengers choosing each class and provides insights into travel preferences among airline customers.



4. Dashboard

Combine all visualizations into a single interactive Power BI Dashboard.

The dashboard can include filters for:

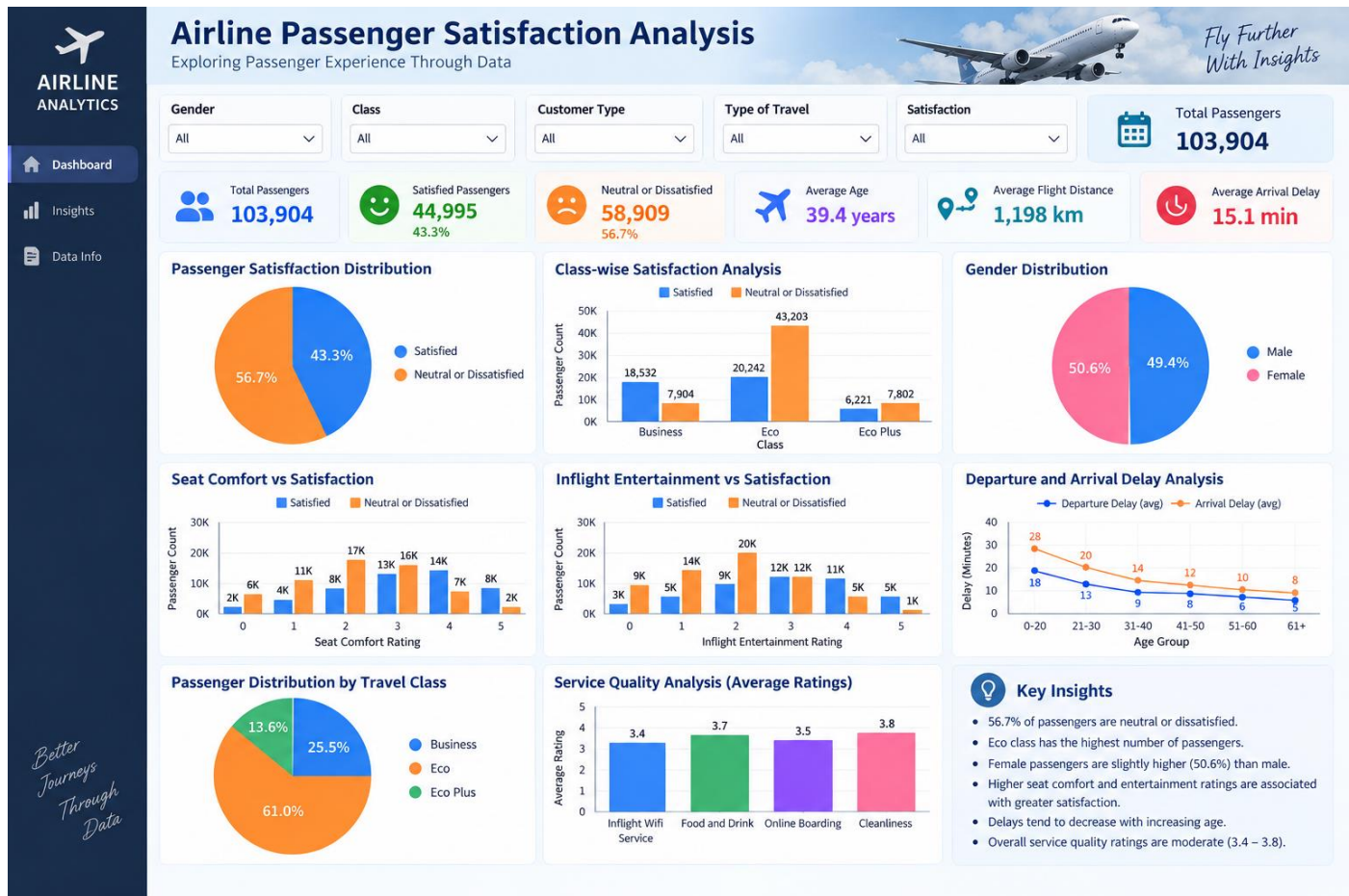
Gender

Class

Customer Type

Type of Travel

Satisfaction



RESULT:

The Airline Passenger Satisfaction Analysis project was successfully completed using Python, SQL Server, and Power BI. A dataset containing 103,904 passenger records was cleaned, stored, analyzed, and visualized to identify trends in passenger satisfaction, service quality, and flight delays.

Final Result

103,904 Records Analyzed

↓

Python Data Processing

↓

SQL Server Storage

↓

Power BI Visualizations

↓

Interactive Dashboard

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Airline Passenger Satisfaction Analysis Completed.